DATABASE DESIGN

Entity, Attribute, Key, Relationship, and Normalization Specification

3.2 Database Design

This document defines the proposed relational database structure for the Web-Based Inventory Application. It identifies the entities, attributes, primary keys, foreign keys, relationships, and normalization approach to serve as the basis for the Entity-Relationship Diagram (ERD) and Data Dictionary.

The structure is based on the project's SRS. Where the SRS does not explicitly define a database structure, the item is identified as a design decision rather than presented as an existing requirement.

3.2.1 Identified Entities

1. User
2. Role
3. Laboratory
4. User_Laboratory
5. Category
6. Inventory_Item
7. Asset
8. Inventory_Transaction
9. Borrowing_Request
10. Borrowing_Request_Item
11. Release
12. Release_Item
13. Return
14. Return_Item
15. Damage_Loss_Record
16. Maintenance_Record
17. Audit_Log

3.2.2 Entity and Attribute Design

User

Attribute	Key	Purpose
user_id	PK	Unique user identifier
role_id	FK	Assigned role
username	UNIQUE	Login identifier
password_hash		Secure password hash
first_name		First name
last_name		Last name

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email		Email, if used
status		Account status
created_at		Creation date/time
updated_at		Last update date/time

Role

Attribute	Key	Purpose
role_id	PK	Unique role identifier
role_name	UNIQUE	Role name
description		Role description

Laboratory

Attribute	Key	Purpose
laboratory_id	PK	Unique laboratory identifier
name		Laboratory name
type		Laboratory type
location		Physical location
description		Description
status		Laboratory status
created_at		Creation date/time

User_Laboratory

Attribute	Key	Purpose
user_id	PK, FK	References User
laboratory_id	PK, FK	References Laboratory

Category

Attribute	Key	Purpose
category_id	PK	Unique category identifier
name	UNIQUE	Category name
description		Description
status		Active/inactive

Inventory_Item

Attribute	Key	Purpose
item_id	PK	Unique inventory identifier
item_name		Item name
description		Description
category_id	FK	Associated category
laboratory_id	FK	Assigned laboratory
item_type		Consumable or equipment
unit_of_measurement		Unit

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quantity		Current quantity
minimum_stock_level		Low-stock threshold
condition		Current condition
location		Storage/location
acquisition_date		Date acquired
acquisition_source		Source
remarks		Additional information
status		Inventory status
created_at		Date added
updated_at		Last update
archived_at		Archive date, if applicable

Asset

Attribute	Key	Purpose
asset_id	PK	Unique asset identifier
item_id	FK	Related inventory item/type
laboratory_id	FK	Current laboratory
status		Operational status
condition		Physical condition
remarks		Additional information
created_at		Registration date
updated_at		Last update

Inventory_Transaction

Attribute	Key	Purpose
transaction_id	PK	Unique transaction
item_id	FK	Affected item
user_id	FK	User performing transaction
transaction_type		Stock-in, stock-out, adjustment, or transfer
quantity		Quantity involved
transaction_date		Date/time
reason		Reason
source		Incoming source, if applicable
remarks		Additional information

Borrowing_Request

Attribute	Key	Purpose
request_id	PK	Unique request
requester_id	FK	Submitting user
laboratory_id	FK	Laboratory involved
purpose		Purpose
request_date		Request date
expected_return_date		Expected return

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status		Workflow status
approved_by	FK	Approver
approval_date		Approval/rejection date
approval_reason		Approval/rejection reason
remarks		Additional information

Borrowing_Request_Item

Attribute	Key	Purpose
request_item_id	PK	Unique request detail
request_id	FK	Related request
item_id	FK	Requested item
requested_quantity		Requested quantity
remarks		Additional information

Release

Attribute	Key	Purpose
release_id	PK	Unique release
request_id	FK	Related request
released_by	FK	Releasing user
release_date		Release date/time
remarks		Additional information

Release_Item

Attribute	Key	Purpose
release_item_id	PK	Unique release detail
release_id	FK	Related release
item_id	FK	Released item
asset_id	FK, nullable	Individual asset, when applicable
quantity_released		Quantity released
remarks		Additional information

Return

Attribute	Key	Purpose
return_id	PK	Unique return
request_id	FK	Related request
processed_by	FK	Processing user
return_date		Return date
remarks		Additional information

Return_Item

Attribute	Key	Purpose

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return_item_id	PK	Unique return detail
return_id	FK	Related return
item_id	FK	Returned item
asset_id	FK, nullable	Individual asset, when applicable
quantity_returned		Quantity returned
missing_quantity		Quantity missing
damaged_quantity		Quantity damaged
condition		Condition upon return
remarks		Additional information

Damage_Loss_Record

Attribute	Key	Purpose
damage_loss_id	PK	Unique damage/loss record
item_id	FK, nullable	Affected inventory item
asset_id	FK, nullable	Affected asset
record_type		Damage or loss
date_recorded		Date recorded
description		Description
reported_by	FK	Reporting user
current_condition		Current condition
action_taken		Action taken
remarks		Additional information

Maintenance_Record

Attribute	Key	Purpose
maintenance_id	PK	Unique maintenance record
asset_id	FK	Maintained asset
reported_issue		Reported issue
date_reported		Report date
maintenance_action		Work performed
responsible_person		Responsible person
date_completed		Completion date
maintenance_status		Status
remarks		Additional information

Audit_Log

Attribute	Key	Purpose
audit_id	PK	Unique audit record
user_id	FK	User performing action
timestamp		Date/time
action		Action performed
affected_entity		Affected entity
record_id		Affected record ID

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previous_value		Previous value, where applicable
new_value		New value, where applicable
reason		Reason, where applicable

3.2.3 Primary Keys

Entity	Primary Key
User	user_id
Role	role_id
Laboratory	laboratory_id
User_Laboratory	user_id + laboratory_id
Category	category_id
Inventory_Item	item_id
Asset	asset_id
Inventory_Transaction	transaction_id
Borrowing_Request	request_id
Borrowing_Request_Item	request_item_id
Release	release_id
Release_Item	release_item_id
Return	return_id
Return_Item	return_item_id
Damage_Loss_Record	damage_loss_id
Maintenance_Record	maintenance_id
Audit_Log	audit_id

3.2.4 Foreign Keys

Entity	Foreign Key	Referenced Entity	Referenced Key
User	role_id	Role	role_id
User_Laboratory	user_id	User	user_id
User_Laboratory	laboratory_id	Laboratory	laboratory_id
Inventory_Item	category_id	Category	category_id
Inventory_Item	laboratory_id	Laboratory	laboratory_id
Asset	item_id	Inventory_Item	item_id
Asset	laboratory_id	Laboratory	laboratory_id
Inventory_Transaction	item_id	Inventory_Item	item_id
Inventory_Transaction	user_id	User	user_id
Borrowing_Request	requester_id	User	user_id
Borrowing_Request	laboratory_id	Laboratory	laboratory_id
Borrowing_Request	approved_by	User	user_id
Borrowing_Request_Item	request_id	Borrowing_Request	request_id
Borrowing_Request_Item	item_id	Inventory_Item	item_id

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Release	request_id	Borrowing_Request	request_id
Release	released_by	User	user_id
Release_Item	release_id	Release	release_id
Release_Item	item_id	Inventory_Item	item_id
Release_Item	asset_id	Asset	asset_id
Return	request_id	Borrowing_Request	request_id
Return	processed_by	User	user_id
Return_Item	return_id	Return	return_id
Return_Item	item_id	Inventory_Item	item_id
Return_Item	asset_id	Asset	asset_id
Damage_Loss_Record	item_id	Inventory_Item	item_id
Damage_Loss_Record	asset_id	Asset	asset_id
Damage_Loss_Record	reported_by	User	user_id
Maintenance_Record	asset_id	Asset	asset_id
Audit_Log	user_id	User	user_id

3.2.5 Relationships

Entity A	Cardinality	Entity B	Relationship
Role	1 : N	User	One role can be assigned to many users.
User	1 : N	User_Laboratory	A user can have multiple laboratory access records.
Laboratory	1 : N	User_Laboratory	A laboratory can have multiple authorized users.
Laboratory	1 : N	Inventory_Item	A laboratory can contain many inventory items.
Category	1 : N	Inventory_Item	One category can contain many inventory items.
Inventory_Item	1 : N	Asset	One inventory item/type may correspond to multiple individually tracked assets.
Inventory_Item	1 : N	Inventory_Transaction	One inventory item can have many inventory transactions.
User	1 : N	Inventory_Transaction	One user can perform many inventory transactions.
User	1 : N	Borrowing_Request	A user can submit many borrowing requests.

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Laboratory	1 : N	Borrowing_Request	A laboratory can be associated with many borrowing requests.
Borrowing_Request	1 : N	Borrowing_Request_Item	One request can contain multiple requested item records.
Inventory_Item	1 : N	Borrowing_Request_Item	An inventory item can appear in many request detail records.
Borrowing_Request	1 : N	Release	A request can be associated with release records; exact multiplicity depends on the final workflow.
Release	1 : N	Release_Item	One release can contain multiple released item records.
Borrowing_Request	1 : N	Return	A request can be associated with return records; exact multiplicity depends on the final workflow.
Return	1 : N	Return_Item	One return can contain multiple returned item records.
Asset	1 : N	Maintenance_Record	An asset can have multiple maintenance records.
User	1 : N	Audit_Log	One user can generate many audit log records.

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3.2.6 Normalization

First Normal Form (1NF)

The design avoids repeating groups and multi-valued fields. Each attribute is intended to contain a single value. Laboratory access is stored in User_Laboratory instead of storing several laboratory names in one User record. Requested items are stored in Borrowing_Request_Item instead of placing multiple items in one request field.

Second Normal Form (2NF)

The design separates attributes that belong to different entities and avoids partial dependency on composite keys. User_Laboratory contains the user-laboratory relationship without duplicating user or laboratory details. Borrowing_Request_Item stores request-specific item details without duplicating requester or item master data.

Third Normal Form (3NF)

The design separates non-key information into the entity to which it belongs. Category details are stored in Category rather than repeated in Inventory_Item. Laboratory details are stored in Laboratory rather than repeated in every related record. This reduces duplication and supports consistent updates.

3.2.7 Normalized Structure

- Role → User
- User ↔ Laboratory through User_Laboratory
- Laboratory → Inventory_Item
- Category → Inventory_Item
- Inventory_Item → Asset
- Inventory_Item → Inventory_Transaction
- User → Inventory_Transaction
- User → Borrowing_Request
- Borrowing_Request ↔ Inventory_Item through Borrowing_Request_Item
- Borrowing_Request → Release → Release_Item
- Borrowing_Request → Return → Return_Item
- Asset → Maintenance_Record
- User → Audit_Log
- Inventory_Item / Asset → Damage_Loss_Record

3.2.8 Design Decisions and Items Requiring Confirmation

Item	Design Note
Multiple roles per user	The SRS defines roles but does not explicitly state whether one user may have multiple roles. The current proposal uses one primary role per user through User.role_id.
Shared inventory	The SRS allows shared inventory but does not define how it should be represented in the database. The final implementation should decide whether a shared item may have a NULL laboratory_id or whether another structure is required.
Asset relationship	The SRS distinguishes individually tracked assets from quantity-based inventory. The proposed design relates Asset to Inventory_Item through item_id.
Multiple releases	The SRS distinguishes approval from physical release but does not explicitly state whether one request may have multiple release events. The final workflow should determine the exact cardinality.
Multiple returns	The SRS requires return information but does not

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	explicitly define whether partial or multiple return events are allowed. The final workflow should determine the exact cardinality.
Damage/loss target	The SRS refers to an item or asset. The proposed Damage_Loss_Record permits either item_id or asset_id and should enforce that exactly one target is provided.
User profile fields	First name, last name, and email are proposed implementation fields because the SRS does not fully define the user profile schema.

3.2.9 ERD Preparation

The entities, keys, and relationships in this document shall be used as the basis for the project's ERD. The ERD should show each entity, its primary key, relevant foreign keys, major attributes, and relationship cardinalities. The ERD and Data Dictionary should remain consistent with the finalized database structure.

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